Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

Analysis I (MTH 301)

Quiz 2

Date: 07/09/2024

Time: 30 minutes | Total marks: 8

Name:	Marks obtained ↓
Roll number:	

Answer to any question must be written within the space provided below the same. Extra sheets can be used for rough works only.

1. Prove or disprove the following statement: let (X, d_X) and (Y, d_Y) are metric spaces and $f : X \longrightarrow Y$. Suppose that $A \subseteq X$. Denote the restriction of f to A by $f|_A$ and consider $f|_A$ as a map from the metric space A (equipped with the induced metric) to the metric space Y. Then the following are equivalent:

   (I) f is continuous at each point of A.

   (II) $f|_A$ is continuous.

[3]

The above statement is false. Consider

$$f : [-1,1] \to \mathbb{R}, \quad f(x) := \begin{cases} -1 , & -1 \le x \le 0 \\ 1 , & 0 < x \le 1. \end{cases}$$

It is clear that f is not continuous at 0, however the restriction of f to $\{0\}$ is constant, and hence continuous. ∎

[SEE PAGE 3 OF THE FILE, TITLED "ANSWERS TO THE DOUBTS" THAT I HAVE SENT TO ALL OF YOU LAST EVENING]

REMARK: There are many more such counter-examples. Indeed, some of the examples which we have already seen in the class will also work.

2. Prove or disprove the following statement: let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$ and $f : A \longrightarrow Y$. Assume that $a \in X$ is a limit point of A and $\lim_{x \to a} f(x) = \ell \, (\in Y)$. Then ℓ is a limit point of $\{f(x) : x \in A\}$. [2]

The above statement is false. Consider $f : \mathbb{R} \to \mathbb{R}$, $f(x) := 0$, $\forall x \in \mathbb{R}$. Then clearly $\{f(x) : x \in \mathbb{R}\} = \{0\}$, which does not have a limit point. Hence, in particular, although $\lim_{x \to 0} f(x) = 0$, 0 is not a limit point of $\{f(x) : x \in \mathbb{R}\}$. ∎

3. Find all subsequential limits of the sequence $\left\{\left(n\left|\sin\frac{n\pi}{3}\right|\right)^{\frac{1}{n}}\right\}_{n=1}^{\infty}$.

[3]

SOLUTION:- We observe that

$$\left(n\left|\sin\frac{n\pi}{3}\right|\right)^{\frac{1}{n}} = \begin{cases} 0 & \text{if } n=3k,\ k\in\mathbb{Z}_{\geq 0} \\ (6k+1)^{\frac{1}{6k+1}}\left|\sin\pi/3\right|^{\frac{1}{6k+1}} & \text{if } n=6k+1, \\ (6k+2)^{\frac{1}{6k+2}}\left|\sin 2\pi/3\right|^{\frac{1}{6k+2}} & " \ n=6k+2, " \\ (6k+4)^{\frac{1}{6k+4}}\left|\sin 4\pi/3\right|^{\frac{1}{6k+4}} & " \ n=6k+4 " \\ (6k+5)^{\frac{1}{6k+5}}\left|\sin 5\pi/3\right|^{\frac{1}{6k+5}} & " \ n=6k+5, " \end{cases}$$

Since $\sin\pi/3$, $\sin 2\pi/3$, $\sin 4\pi/3$ and $\sin 5\pi/3 \neq 0$, it follows that,

$$\left|\sin r\pi/3\right|^{\frac{1}{6k+r}} \xrightarrow[k\to\infty]{} 1 \quad \text{for } r=1,2,4,5. \quad \cdots (i)$$

On the other hand, ~~it~~ we have $n^{1/n} \xrightarrow[n\to\infty]{} 1 \cdots (ii)$

It now follows from (i) & (ii) that,

$$(6k+r)^{\frac{1}{6k+r}}\left|\sin(6k+r)\pi/3\right|^{\frac{1}{6k+r}} \xrightarrow[k\to\infty]{} \begin{cases} 0 & r=0,3 \\ 1 & r=1,2,4,5 \end{cases}$$

As $\mathbb{N}$ is the disjoint union of the following subsets: $\{6k+r : k=0,1,2,\cdots\}$, $r=0,1,\ldots,5$ we conclude that 0 and 1 are the only subsequential limits. ∎